

TONY THOMAS

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EDUCATION

Indraprastha Institute of Information Technology Delhi

B. Tech in Computer Science and Design

Dec 2021 - May 2025

CGPA: 8.1/10

WORK EXPERIENCE

National Australia Bank | Aug 2025 - Present

Associate Software Engineer

- Built an automated service-communication microservice for investment events (buy/rebalance/withdrawals) using **Spring Boot**, JPA, WebClient, and batch schedulers, reducing manual operations by 80% and improving customer clarity.
- Led frontend architecture and UI/UX uplift across multiple micro-frontends, enforcing clean component boundaries and scalable patterns with **React + TypeScript**, improving delivery velocity by 30%.
- Implemented micro-frontend and **GraphQL** integrations for product origination and target-market determination flows, collaborating on schema and API design to enable parallel frontend-backend development and reduce integration defects.
- Established end-to-end test automation and CI/CD, replicating production scenarios with **Docker**, frontend and backend automation (**React testing**, **AutoX**, **JUnit**), and Harness/Jenkins pipelines, cutting regressions by 40%.

Infraprime Logistics Technologies Pvt. Ltd | Jun 2024 - Dec 2024

Software Intern

- Designed and owned an end-to-end GPS tracking and route optimization system using **Python**, **FastAPI**, **Raspberry Pi** (Raspberry Pi OS), and Mappls APIs, improving route efficiency by ~40% across 250+ vehicles and enabling real-time trip visibility.
- Implemented edge-based **ML inference** on Raspberry Pi using **TensorFlow**, accelerometer and sensor data, and linear regression-based models to detect unsafe driving behavior, triggering real-time alerts and reducing risky driving incidents by ~20%.
- Built a **real-time data ingestion and analytics** pipeline using **SQLite**, **InfluxDB**, and **Apache Kafka** for sensor and trip data, processing 10,000+ daily events with reliable cloud sync and low-latency streaming.

Shashwat Devalay(ADS Lab) | Jan 2024 - May 2024

Full Stack Developer

- Designed and built a cross-platform community service mobile app using **React Native**, **Expo**, **React Hooks**, and **Firebase**, connecting volunteers and service seekers through scalable, real-time workflows.
- Owned the admin and event management system, implementing automated volunteer enrollment, event approvals, and moderation via a custom **Admin Dashboard**, reducing administrative effort by ~40%.
- Implemented **real-time chat with WebSocket** and Swarozgar marketplace modules, enabling secure messaging and streamlined shop registration/approvals, supporting 1,000+ users and empowering 200+ small businesses/artisans.

PROJECTS

E-Commerce Microservices Platform

Java, Spring Boot, PostgreSQL, Flyway, Swagger UI, JWT, Junit, Maven

[GitHub](#)

- Architected REST APIs with **Spring Boot**, JWT authentication, and MVC pattern, handling product management, shopping cart operations, and order processing
- Implemented payment processing with Stripe integration, webhook handling, and **Flyway** database migrations, ensuring data consistency and transaction reliability
- Containerized services using Docker and set up CI/CD pipeline with automated testing (JUnit, Mockito), achieving 95% test coverage.
- Developed comprehensive API documentation with **Swagger UI** and implemented **role-based access control** for admin/customer roles.

Weather Monitoring and Drought Assessment for Farmers

Python, SQL, MySQL, Hadoop, Apache Spark, Tensorflow

[GitHub](#)

- Built data warehouse processing 3M+ agricultural records with automated **ETL pipelines**, ensuring 99.9% data reliability and fault tolerance
- Developed **Spark Streaming** analytics pipeline delivering real-time weather forecasts, drought risk alerts, and crop recommendations to 1,500+ farmers via mobile/web interfaces
- Implemented **predictive models** using time-series analysis, reducing crop loss risks by 25% through early warning systems.

Smart News Mobile Application

Kotlin, Android Studio, Jetpack Compose, Room DB, Retrofit

[GitHub](#)

- Built Android app using Jetpack Compose with location-aware news delivery, increasing local news engagement by 40% through personalized content curation
- Implemented offline-first architecture using Room Database and Retrofit, enabling news consumption without internet and boosting user retention by 25%

SKILLS

Programming Languages: Java, TypeScript, Python, SQL, C++, HTML5/CSS, Kotlin

Frontend: React.js, Apollo GraphQL (Client), Redux.js, Material UI, TailwindCSS, Vite, Jest, Cypress, AutoX

Backend: Spring Boot (Security, Data JPA, Web MVC, Batch), RESTful APIs, Swagger, JWT, Gradle, Maven, JUnit 5, Mockito

Databases: PostgreSQL, MongoDB, Redis, Neo4J, Apache Kafka, Flyway

Tools & Platforms: IntelliJ IDEA, Visual Studio Code, GitHub, DBeaver, Postman, Splunk, OpenSearch

Cloud & DevOps: AWS (Lambda, S3, EC2), Docker, Jenkins, Harness, HashiCorp Vault

LEADERSHIP AND COMMUNITY

- Teaching Assistant in Human Centered Interaction and Design of Interactive Systems.
- Held many leadership roles in CyFuse, BioBytes and IEEE club.
- Founding Member of Fresources.